cs4765-project

May 1, 2025

[ ]:
```
# use pandas to load data and visualize
# use longformer tokenizer (efficient at tokenizing large texts)
# Set Up the Model for Fine-Tuning (transition to Colab here?)
# Define Training Arguments
from google.colab import drive
drive.mount('/content/drive')
#!pip install -r '/content/drive/My Drive/CS4765 Datasets/requirements.txt'
```

[ ]:
```
# was having a lot of trouble making the requirements.txt file work in Colab
# so I am manually adding them here as a last resort
# IMPORTANT: If you are using jupyter notebook, please ensure these libraries␣
 ↪are installed!
!pip install datasets transformers torch matplotlib seaborn numpy
#!pip install -r '/content/drive/My Drive/CS4765 Datasets/requirements.txt'
```

[ ]:
```
# Visualizing the data with pandasimport pandas as pd
# This cell was used for testing in the colab environment, where we were able␣
 ↪to mount our google drive
import pandas as pd
import json

# Ths commented out line below is what was used in the colab environment
# data = pd.read_json('/content/drive/My Drive/CS4765 Datasets/
 ↪subtaskA_train_monolingual.jsonl', lines=True)

# this line is for running on jupyter notebook, assuming you have the datasets␣
 ↪stored in a datasets folder within the root
data = pd.read_json('./datasets/subtaskA_train_monolingual.jsonl', lines=True)
print(type(data))
print(data.head())

# visualizing the data, running this slows the computer down a lot because␣
 ↪there's a lot of data beware!
# for index, row in data.iterrows():
#     print(f"Row {index}: {row}")
```

[ ]:
```
# Some random testing with tokenizer
# # tokenizer function
from transformers import LongformerTokenizer


tokenizer = LongformerTokenizer.from_pretrained("allenai/longformer-base-4096")
print(tokenizer("Hello world")["input_ids"])
print(tokenizer('./datasets/subtaskA_dev_monlingual.jsonl')["input_ids"])
def tokenize_function(examples):
    return tokenizer(examples['text'], padding='max_length', truncation=True,␣
 ↪max_length=4096)
```

[ ]:
```
# Testing the tokenizer on our training dataset
from datasets import Dataset

dataset = Dataset.from_pandas(data)
tokenized_dataset = dataset.map(tokenize_function, batched=True)
print(tokenized_dataset)
```

[ ]:
```
# Testing loading dataset with hugging face library
from datasets import load_dataset

# Loading the dataset from google drive (for colab environment)
# raw_datasets = load_dataset('json',data_files='/content/drive/My Drive/CS4765␣
 ↪Datasets/subtaskA_train_monolingual.jsonl')
# rawvalidation_dataset = load_dataset('json',data_files='/content/drive/My␣
 ↪Drive/CS4765 Datasets/validation.jsonl')

# these line below is for running the dataset within jupyter notebook assuming␣
 ↪that the dataset is stored in a datasets folder within root directory
raw_datasets = load_dataset('json',data_files='./datasets/
 ↪subtaskA_train_monolingual.jsonl') # load train set
rawvalidation_dataset = load_dataset('json',data_files='./datasets/validation.
 ↪jsonl') # load validation set

# Some poking around the raw dataset to see how it got formatted by hugging␣
 ↪face data loader
print(raw_datasets['train'][:5])
print(rawvalidation_dataset['train'][:5])
tokenized_subset = raw_datasets["train"].select(range(5)).
 ↪map(tokenize_function, batched=True)
print(tokenized_subset)
```

[ ]:
```
# This cell was supposed to train a longformer model
# We ran into issues with training this model because of how much compute power␣
 ↪was necessary
# We tried lowering the max position from 4096 to 2048 but kept timing out
```

```
# We also tried running a stronger GPU within Google Colab but were not␣
 ↪succesful
from transformers import AutoTokenizer, AutoModelForSequenceClassification,␣
 ↪Trainer, TrainingArguments, LongformerConfig,␣
 ↪LongformerForSequenceClassification
from datasets import load_dataset
import os

os.environ["WANDB_MODE"] = "disabled"  # Disable wandb mode because it keeps␣
 ↪asking for an API key

# Load raw dataset
# Please make sure to adjust the path as needed if you want to try training the␣
 ↪longformer
# The path below was for running in a Google Colab environment where my google␣
 ↪drive was mounted
# I do not recommend running this on a local machine as it is very resource␣
 ↪intensive
raw_datasets = load_dataset('json', data_files='/content/drive/My Drive/CS4765␣
 ↪Datasets/subtaskA_train_monolingual.jsonl')  # For training
# For validation while training, we took about 20% of the train data and put it␣
 ↪into a separate file
rawvalidation_dataset = load_dataset('json', data_files='/content/drive/My␣
 ↪Drive/CS4765 Datasets/validation.jsonl')

model_name = "allenai/longformer-base-4096"

tokenizer = AutoTokenizer.from_pretrained(model_name, model_max_length=2048)

config = LongformerConfig.from_pretrained(model_name)
config.max_position_embeddings = 2048 # Adjust model configuration for 2048 max␣
 ↪position embeddings, to hopefully increase training speed
config.num_labels = 2

model = LongformerForSequenceClassification.from_pretrained(model_name,␣
 ↪config=config)

# Tokenize function using Longformer tokenizer
def tokenize_function(examples):
    return tokenizer(examples["text"], padding="max_length", truncation=True)

# Tokenize the datasets with above function
tokenized_dataset_train = raw_datasets["train"].map(tokenize_function,␣
 ↪batched=True)
tokenized_validation = rawvalidation_dataset["train"].map(tokenize_function,␣
 ↪batched=True)
```

```
# Define training arguments with default values
training_args = TrainingArguments(
    output_dir="./results",
    evaluation_strategy="epoch",
    save_strategy="epoch",
    save_steps=100,
    learning_rate=2e-5,
    per_device_train_batch_size=4,
    num_train_epochs=3,
    weight_decay=0.01,
    save_total_limit=1,
    logging_dir="./logs",
    load_best_model_at_end=True,
    log_level="debug",
)

# Create Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_dataset_train,
    eval_dataset=tokenized_validation,
)

# Train the model in try catch to make sure we catch silent errors
try:
    trainer.train()  # Add `resume_from_checkpoint=True` if training stops␣
 ↪unexpectedly
except Exception as e:
    print(f"Training error: {e}")

# Save the model and tokenizer
print("Saving model...")
model.save_pretrained("./longformer_model")
print("Model saved.")

print("Saving tokenizer...")
tokenizer.save_pretrained("./longformer_model")
print("Tokenizer saved.")
```

[ ]:
```
# This cell trains a distilbert model
# the code is very similar to above cell
from transformers import AutoTokenizer, AutoModelForSequenceClassification,␣
 ↪Trainer, TrainingArguments
from datasets import load_dataset
import os
```

```
import torch


os.environ["WANDB_MODE"] = "disabled"  # Disable wandb mode

# Load raw dataset
# As mentioned before, please adjust the filepath to ther data_files as␣
 ↪necessary
raw_datasets = load_dataset('json', data_files='/content/drive/My Drive/CS4765␣
 ↪Datasets/subtaskA_train_monolingual.jsonl')
rawvalidation_dataset = load_dataset('json', data_files='/content/drive/My␣
 ↪Drive/CS4765 Datasets/validation.jsonl')

# Load distilbert tokenizer and model
model_name = "distilbert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSequenceClassification.from_pretrained(model_name,␣
 ↪num_labels=2)

# Tokenize the dataset using DistilBERT tokenizer
def tokenize_function(examples):
    return tokenizer(examples["text"], padding="max_length", truncation=True)

tokenized_dataset_train = raw_datasets["train"].map(tokenize_function,␣
 ↪batched=True)
tokenized_validation = rawvalidation_dataset["train"].map(tokenize_function,␣
 ↪batched=True)

# Define training arguments
training_args = TrainingArguments(
    output_dir="./results",
    evaluation_strategy="epoch",
    save_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
    save_total_limit=1,
    logging_dir="./logs",
    load_best_model_at_end=True,
    log_level="debug",
)

# Create Trainer
trainer = Trainer(
    model=model,
```

```
    args=training_args,
    train_dataset=tokenized_dataset_train,
    eval_dataset=tokenized_validation,
)

# Train the model
try:
    trainer.train()
except Exception as e:
    print(f"Training error: {e}")

# Save it so we don't need to retrain
print("Saving model...")
model.save_pretrained("./distilbert_model")
print("Model saved.")

print("Saving tokenizer...")
tokenizer.save_pretrained("./distilbert_model")
print("Tokenizer saved.")
```

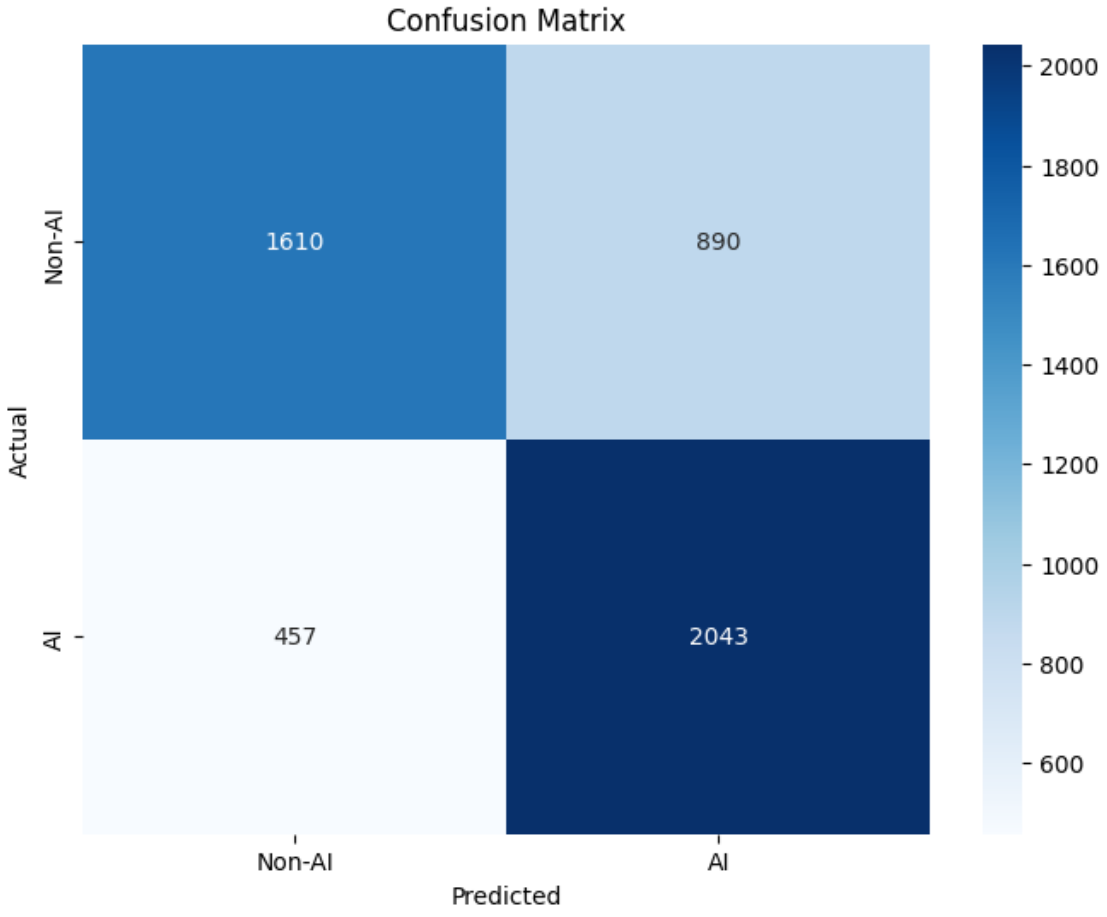
[ ]:
```
# unzip model, run this to unzip the zipfile of the model
# IMPORTANT: this assumes that the zip file is stored in the root directory
# THIS HAS TO BE RUN FIRST BEFORE RUNNING THE MODEL
import sys
from zipfile import PyZipFile
pzf = PyZipFile('./distilbert_model.zip')
pzf.extractall()
```

[2]:
```
# If we wnat to reuse the model we trained we can use this code now
# PLEASE MAKE SURE THE ZIP FILE WAS UNZIPPED WITH THE CELL ABOVE
# This cell uses our test data on our fine tuned distilled bert model
# !!THIS CELL TAKES TIME TO RUN!!

from transformers import Trainer, AutoTokenizer,␣
 ↪AutoModelForSequenceClassification
from sklearn.metrics import confusion_matrix, classification_report
from datasets import load_dataset

import seaborn as sns
import torch

# Load the saved model and tokenizer
model = AutoModelForSequenceClassification.from_pretrained("./distilbert_model")
tokenizer = AutoTokenizer.from_pretrained("./distilbert_model")

# Load a test dataset
```

```
# raw_test_dataset = load_dataset('json', data_files='/content/drive/MyDrive/
 ↪CS4765 Datasets/subtaskA_dev_monolingual.jsonl')
# Below code is for running in jupyter notebook assuming dataset is inside a␣
 ↪datasets folder in root directory
raw_test_dataset = load_dataset('json', data_files='./datasets/
 ↪subtaskA_dev_monolingual.jsonl')

# Tokenize the test dataset
def tokenize_function(examples):
    return tokenizer(examples["text"], padding="max_length", truncation=True)

tokenized_test = raw_test_dataset["train"].map(tokenize_function, batched=True)



# Move model to device (GPU/CPU)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)

# function to make predictions
def get_predictions(model, dataset):
    model.eval()
    predictions = []
    true_labels = []
    with torch.no_grad():
        for example in dataset:
            input_ids = torch.tensor([example["input_ids"]]).to(device)
            attention_mask = torch.tensor([example["attention_mask"]]).
 ↪to(device)
            label = example["label"]

            outputs = model(input_ids=input_ids, attention_mask=attention_mask)
            logits = outputs.logits
            predicted_label = torch.argmax(logits, dim=1).item()

            predictions.append(predicted_label)
            true_labels.append(label)
    return predictions, true_labels

# Get predictions and true labels
predictions, true_labels = get_predictions(model, tokenized_test)
```

```
/opt/homebrew/Cellar/jupyterlab/4.2.5_1/libexec/lib/python3.12/site-
packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update
jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

[3]:
```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix
import seaborn as sns
# Create the confusion matrix
cm = confusion_matrix(true_labels, predictions)

# use matplotlib to visualize the confusion matrix
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=["Non-AI",␣
 ↪"AI"], yticklabels=["Non-AI", "AI"])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

print(classification_report(true_labels, predictions, target_names=["Non-AI",␣
 ↪"AI"]))
```



```
              precision    recall  f1-score   support

      Non-AI       0.78      0.64      0.71      2500
          AI       0.70      0.82      0.75      2500

    accuracy                           0.73      5000
   macro avg       0.74      0.73      0.73      5000
weighted avg       0.74      0.73      0.73      5000
```

[4]:
```
# This cell will compare our calculations above with a most frequent class as a␣
 ↪baseline
from sklearn.metrics import classification_report, confusion_matrix
import numpy as np


ai_count = true_labels.count(1)  # Count occurrences of label 1
human_count = true_labels.count(0)  # Count occurrences of label 0

if ai_count > human_count:
    mfc = 1
else:
    mfc = 0

print(f"Most Frequent Class: {mfc}")

# Generate baseline predictions (all examples are predicted as the most␣
 ↪frequent class)
baseline_predictions = [mfc] * len(true_labels)

# Step 3: Evaluate the baseline
print("Baseline Classification Report:")
print(classification_report(true_labels, baseline_predictions, zero_division=0))

# Confusion matrix
baseline_cm = confusion_matrix(true_labels, baseline_predictions)
```

```
Most Frequent Class: 0
Baseline Classification Report:
              precision    recall  f1-score   support

           0       0.50      1.00      0.67      2500
           1       0.00      0.00      0.00      2500

    accuracy                           0.50      5000
   macro avg       0.25      0.50      0.33      5000
weighted avg       0.25      0.50      0.33      5000
```

[5]:
```
# We can also use this code for simple testing with one text
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch

def tokenize_function(examples):
    return tokenizer(examples["text"], padding="max_length", truncation=True)

# Load the saved model and tokenizer
model = AutoModelForSequenceClassification.from_pretrained("./distilbert_model")
tokenizer = AutoTokenizer.from_pretrained("./distilbert_model")



def classifyText(example_text):
    # Tokenize the input
    inputs = tokenizer(example_text, return_tensors="pt", padding="max_length",␣
 ↪truncation=True)

    # Move tensors to GPU, this line is only relevent for running on Colab
    # device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    # model = model.to(device)
    inputs = {key: value.to(device) for key, value in inputs.items()}

    # Perform prediction
    outputs = model(**inputs) # ** automatically unloads a dictionary in␣
 ↪python, cool!
    logits = outputs.logits
    predicted_class = torch.argmax(logits, dim=1).item()

    print(f"Predicted Class: {predicted_class}")
```

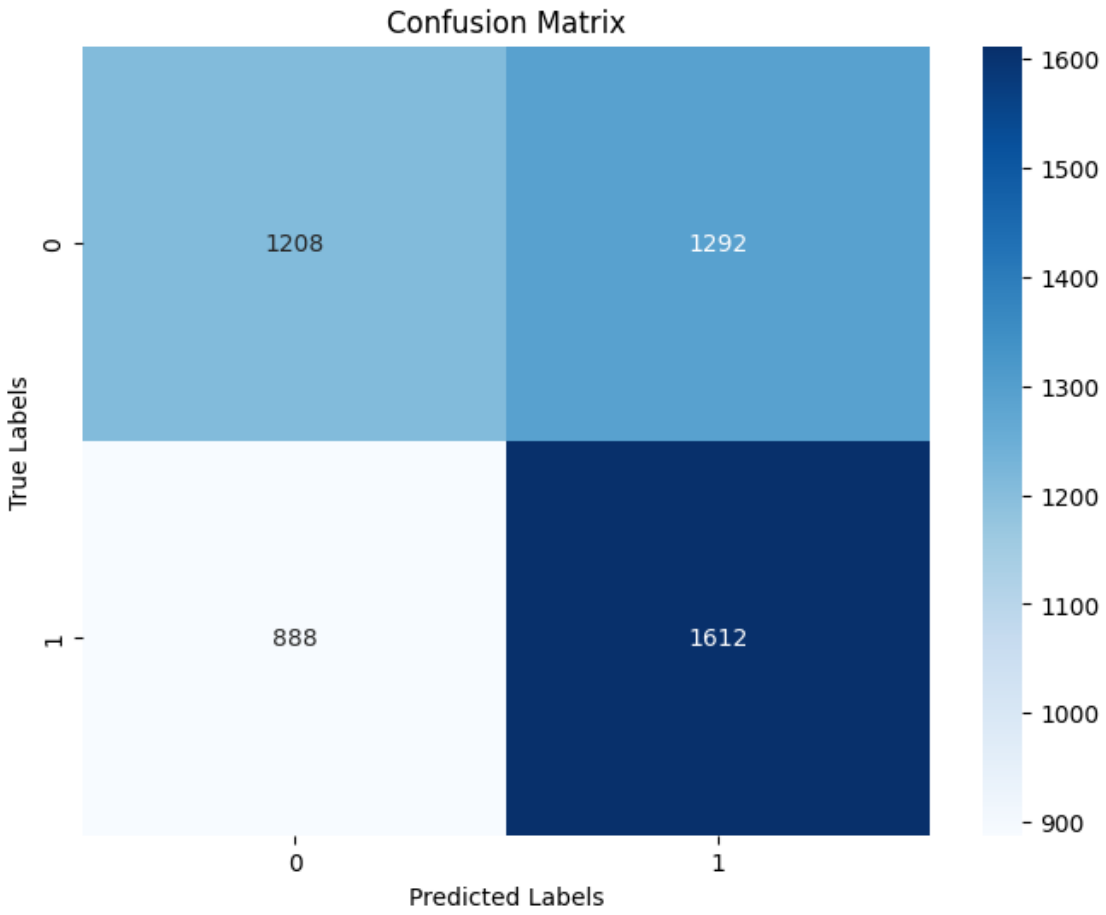
[6]:
```
# Here are 2 short texts to test against the classifier
# The model is classifying both texts as AI which is wrong.
# We believe that the length of text is a critical factor for the model to␣
 ↪determine if it is AI generated or not
# 1-2 sentences may not be enough for our fine tuned distil bert model to␣
 ↪determing if it is AI generated

text1 = "Artificial intelligence is revolutionizing many industries by enabling␣
 ↪automation and deeper insights through data analysis."
text2 = "Hello Dr. Cook, I wrote this message myself. I really hope that my␣
 ↪classifier doesn't think this was AI"
print('text1 result: ')
classifyText(text1)
print('\ntext2 result: ')
classifyText(text2)
```

```
text1 result:
Predicted Class: 1

text2 result:
Predicted Class: 1
```

[7]:
```python
# This cell trains the logistic regression model with the train dataset and tests it with the dev dataset
# These datasets were the same ones used for training DistilBert
# THIS CELL TAKES TIME TO RUN
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, classification_report
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from datasets import load_dataset
import seaborn as sns
import matplotlib.pyplot as plt

# IMPORTANT: Adjust file paths to be what do you have
train_file_path = "./datasets/subtaskA_train_monolingual.jsonl"
test_file_path = "./datasets/subtaskA_dev_monolingual.jsonl"

# Load raw datasets
train_dataset = load_dataset('json', data_files=train_file_path)
validation_dataset = load_dataset('json', data_files=test_file_path)

# Extract the 'text' and 'label' columns from the loaded datasets
train_texts = train_dataset["train"]["text"]
train_labels = train_dataset["train"]["label"]
validation_texts = validation_dataset["train"]["text"]
validation_labels = validation_dataset["train"]["label"]

# Tokenize the dataset using the custom tokenizer
count_vectorizer = CountVectorizer()
train_counts = count_vectorizer.fit_transform(train_texts)
validation_counts = count_vectorizer.transform(validation_texts)

# Optional : Scale the data (StandardScaler)
scaler = StandardScaler(with_mean=False)
train_counts_scaled = scaler.fit_transform(train_counts)
validation_counts_scaled = scaler.transform(validation_counts)

# Train the model
lr = LogisticRegression(solver='sag', penalty='l2', max_iter=20000, random_state=0)
lr_classifier = lr.fit(train_counts_scaled, train_labels)
```

```
# Generate predictions on the validation set
validation_predictions = lr_classifier.predict(validation_counts_scaled)

# Calculate confusion matrix
conf_matrix = confusion_matrix(validation_labels, validation_predictions)
print("Confusion Matrix:")
print(conf_matrix)

# Plot the confusion matrix
plt.figure(figsize=(8, 6))
sns.heatmap(
    conf_matrix,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=lr_classifier.classes_,
    yticklabels=lr_classifier.classes_
)
plt.xlabel('Predicted Labels')
plt.ylabel('True Labels')
plt.title('Confusion Matrix')
plt.show()

# Print classification report
class_report = classification_report(validation_labels, validation_predictions,␣
 ↪zero_division=0)
print("Classification Report:")
print(class_report)
```

```
Confusion Matrix:
[[1208 1292]
 [ 888 1612]]
```




```
Classification Report:
              precision    recall  f1-score   support

           0       0.58      0.48      0.53      2500
           1       0.56      0.64      0.60      2500

    accuracy                           0.56      5000
   macro avg       0.57      0.56      0.56      5000
weighted avg       0.57      0.56      0.56      5000
```

[ ]:
```
# Util cell to export our trained model from colab
from google.colab import files
!zip -r distilbert_model.zip distilbert_model
files.download('distilbert_model.zip')
```

[ ]:
```

```